

# AI Insight Sprint™ Case Study #1

## Business Intelligence Portfolio Project

### Fashion E-commerce Executive Business Intelligence Dashboard

*Transforming Raw Business Data into Actionable Executive Insights*

#### Prepared By

Joshua Odekina

Data Analyst | Business Intelligence Analyst

#### Tools Used

- Microsoft Power BI
- DAX
- Power Query
- Python
- Git
- GitHub

#### Framework

AI Insight Sprint™

#### Date

August 2026

# Executive Summary

## Project Overview

This project presents an end-to-end Business Intelligence solution designed to evaluate the operational performance of a fashion e-commerce business. Using the AI Insight Sprint™ Framework, multiple datasets were integrated and transformed into an interactive Power BI dashboard that provides executives with actionable insights across revenue performance, customer behaviour, marketing effectiveness and strategic business performance.

The dashboard enables decision-makers to monitor key performance indicators (KPIs), identify business trends and support evidence-based decision-making through interactive reporting.

## Project Statistics

| Metric              | Value |
|---------------------|-------|
| Total Datasets      | 4     |
| Dashboard Pages     | 5     |
| Unique KPI Measures | 11    |
| DAX Measures        | 11    |
| Visualisations      | 31    |

## Business Objectives

The project was developed to:

- Analyse overall business revenue performance.
- Monitor customer acquisition and retention.
- Evaluate marketing campaign effectiveness.
- Identify high-performing customer segments.
- Provide executive-level strategic recommendations.
- Demonstrate practical Business Intelligence capabilities using Microsoft Power BI.

# **Dashboard Components**

The solution consists of five interactive dashboard pages:

## **Executive Dashboard**

Provides an overall business performance summary through key performance indicators including Total Revenue, Total Transactions, Average Order Value, Unique Customers, Revenue per Customer and Repeat Customer Rate.

## **Revenue Intelligence**

Analyses revenue trends across time, sales channels, countries, customer age groups and discount performance.

## **Customer Intelligence**

Explores customer behaviour, repeat purchasing, customer segmentation and revenue contribution.

## **Marketing Intelligence**

Evaluates campaign duration, marketing channels and promotional performance.

## **Executive Insights**

Summarises business findings and provides strategic recommendations for management.

## Key Findings

The analysis revealed several important business insights:

- Revenue reached its highest level during May before declining significantly in June.
- Germany and France generated the largest share of total revenue.
- Customers aged 26–45 represented the highest-value customer segment.
- The E-commerce sales channel outperformed the App Mobile channel in revenue generation.
- Most revenue was generated from transactions without discounts, suggesting strong customer willingness to purchase at standard prices.
- Repeat customer performance indicates opportunities to strengthen customer retention through loyalty initiatives.
- Marketing campaigns remained active for an average of approximately 9 days, with durations ranging from 7 to 12 days.

## Strategic Recommendations

Based on the findings, the following recommendations are proposed:

- Implement customer loyalty programmes to increase repeat purchases.
- Increase investment in high-performing digital marketing channels.
- Focus promotional campaigns on high-value customer segments.
- Expand marketing efforts within top-performing countries.
- Monitor monthly revenue trends to improve sales consistency.
- Continuously evaluate campaign performance using Business Intelligence dashboards.

## Business Impact

The completed dashboard provides management with:

- A centralised Business Intelligence reporting solution.
- Improved visibility into organisational performance.
- Faster executive decision-making.
- Enhanced customer and marketing analysis.
- Actionable recommendations for business growth.

## Business Value Delivered

This Business Intelligence solution enables executives to:

- Monitor business performance in real time.
- Identify high-performing customer segments.
- Measure marketing effectiveness.
- Improve strategic decision-making.
- Support sustainable revenue growth.

## Technologies Used

| Technology                      | Purpose                                                                  |
|---------------------------------|--------------------------------------------------------------------------|
| Microsoft Power BI              | Dashboard development, interactive visualizations and business reporting |
| DAX (Data Analysis Expressions) | Creation of KPIs, calculated measures and business metrics               |
| Power Query                     | Data extraction, transformation and preparation (ETL)                    |
| Python                          | Data preparation, exploratory analysis and validation                    |
| Git                             | Version control and project change management                            |
| GitHub                          | Repository hosting, project documentation and portfolio publication      |

## Skills Demonstrated

- Business Intelligence
- Data Modelling
- Dashboard Design
- Data Storytelling
- Executive Reporting
- KPI Development
- DAX
- Power Query
- Business Analysis
- Strategic Decision Support

## Project Outcome

The AI Insight Sprint™ Fashion E-commerce Dashboard successfully transformed raw operational data into meaningful executive insights. The project demonstrates the practical application of Business Intelligence techniques to solve real-world business problems while delivering measurable value to decision-makers.

## Author

**Joshua Odekina**

*GitHub Repository*

*[https://github.com/insightbyjosh/AI\\_Insight\\_Sprint\\_Fashion\\_Ecommerce](https://github.com/insightbyjosh/AI_Insight_Sprint_Fashion_Ecommerce)*

***Data Analyst | Business Intelligence Analyst***

*Specialising in Business Intelligence,*

*Executive Dashboards,*

*Data Storytelling,*

*Power BI,*

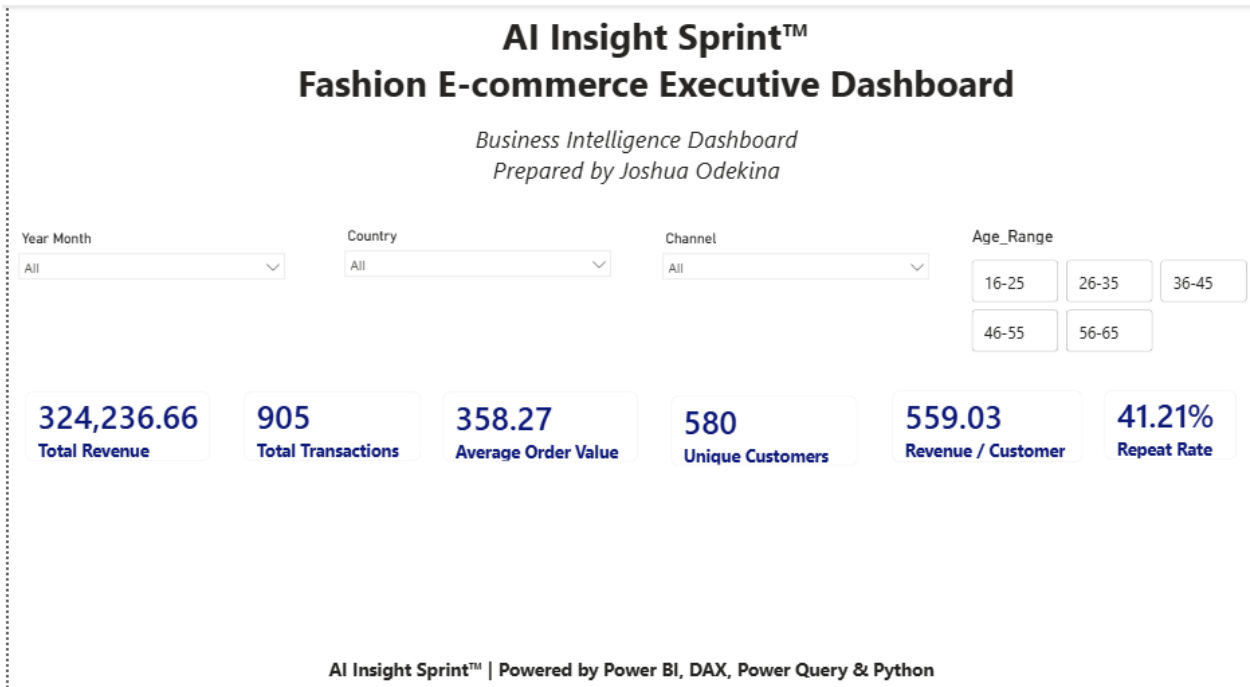
*DAX,*

*and Decision Support Analytics.*

***AI Insight Sprint™***

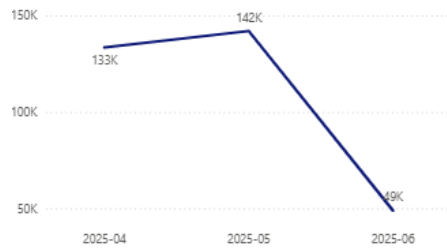
***August 2026***

APPENDIX

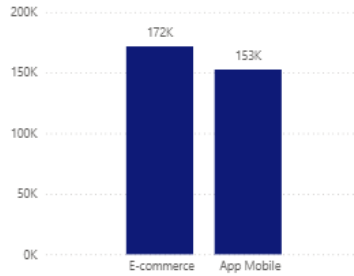


## AI Insight Sprint™ Detailed Business Intelligence Revenue Intelligence

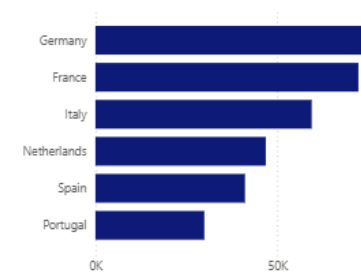
Monthly Revenue Trend



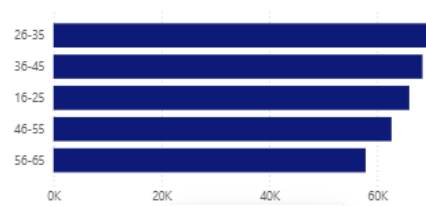
Revenue by Sales Channel



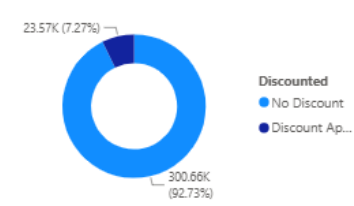
Revenue by Country



Revenue by Age\_Range



Revenue by Discount Status



### Key Revenue Insights

- Revenue peaked in May before declining sharply in June.
- E-commerce generated more revenue than the App Mobile channel.
- Germany and France contributed the largest share of revenue.
- Customers aged 26–35 generated the highest revenue.
- Non-discounted sales accounted for most total revenue.

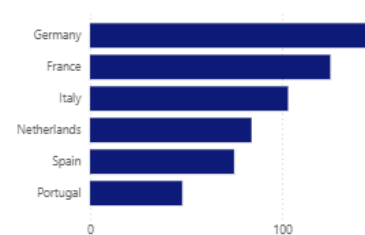
Revenue Intelligence

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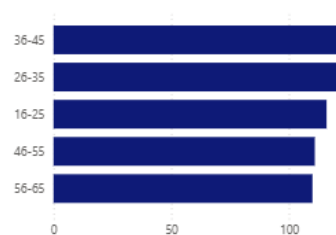
## AI Insight Sprint™ Customer Intelligence

Customer Behaviour & Loyalty Analysis

Customer Distribution by Country



Customer Distribution by Age Group



### Customer Insights

- Germany has the largest customer base, indicating the strongest market penetration among all countries.
- Customers aged 36–45 and 26–35 represent the largest customer segments, highlighting the company's strongest target demographics.
- Approximately 41.21% of customers made repeat purchases, demonstrating moderate customer loyalty with opportunities for improvement.
- Each customer generated an average revenue of 559.03 through approximately 1.56 transactions during the analysis period.
- Strengthening customer retention initiatives and loyalty programmes could increase repeat purchases, improve customer lifetime value, and support sustainable revenue growth.

239

Repeat Customers

341

One-Time Customers

41.21%

Repeat Rate

559.03

Revenue / Customer

1.56

Average Transactions per Customer

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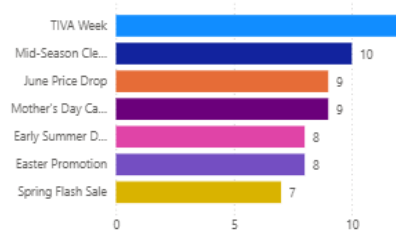


## AI Insight Sprint™

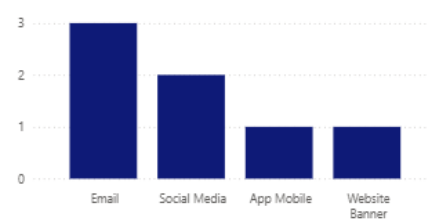
### Marketing Intelligence

Campaign Performance & Marketing Effectiveness

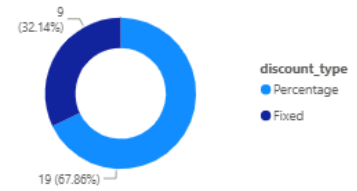
Campaign Duration by Campaign



Campaign Distribution by Channel



Campaigns by Discount Type



9.00

Average Campaign Duration

12

Longest Campaign Duration

#### Marketing Insights

- Campaign durations ranged from 7 to 12 days (average: 9 days), reflecting a strategy of short-term promotional campaigns.
- Marketing activities were distributed across multiple channels, with Email being the most frequently used channel.
- Both percentage and fixed-value discounts were used, providing flexibility in promotional strategies.
- Regular evaluation of channel performance and campaign effectiveness can improve marketing return on investment (ROMI).

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## AI Insight Sprint™

### Executive Insights

Strategic Business Recommendations & Decision Support



#### Executive Summary

The analysis of the fashion e-commerce business revealed valuable insights into revenue performance, customer behaviour, and marketing effectiveness. While the business maintained healthy revenue generation and customer acquisition across multiple markets, opportunities exist to strengthen customer retention, improve campaign effectiveness, and optimise marketing investments. The dashboard highlights clear areas where management can make data-driven decisions to improve profitability, customer lifetime value, and sustainable business growth.



#### Strategic Recommendations

- 1. Increase Customer Retention**  
Introduce loyalty programmes and personalised offers to improve repeat purchase rates beyond the current 41.21%.
- 2. Optimise Marketing Channels**  
Allocate more marketing resources to the highest-performing channels while continuously evaluating campaign effectiveness.
- 3. Focus on High-Value Markets**  
Expand promotional efforts in Germany and France, where revenue and customer acquisition are strongest.
- 4. Target High-Performing Customer Segments**  
Develop personalised campaigns for customers aged 26–45, the company's most valuable customer demographic.
- 5. Improve Revenue Stability**  
Investigate the significant revenue decline observed in June and implement strategies to maintain consistent monthly sales performance.



#### Key Business Metrics

324,236.66  
Total Revenue

580  
Unique Customers

41.21%  
Repeat Rate

358.27  
Average Order Value



#### Expected Business Impact

- Increased customer retention and lifetime value.
- Improved marketing return on investment (ROMI).
- Better allocation of marketing resources.
- Stronger revenue growth across key markets.
- Enhanced executive decision-making through data-driven insight.

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